IIT KHARAGPUR AI4ICPS I HUB FOUNDATION

"Walmart B2B Advanced AI Certification Programme - Cohort 3"

(An Online Certification Programme)

Week	Session Title	Session Date	Start Time (IST)	End Time (IST)	Q&A Time (IST)	Instructor	Prerequisite Video Title
1	Introduction to Generative AI: Tokenizers, Embeddings, Transformer Models, Self-Attention, and Positional Encoding	16 Sep	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Dr. Dr. Jiaul Hoque Paik (IIT KGP)	- Advanced Deep Learning (Theory) (Transformers, BERT)
1	Hands-on Lab: Running pre-trained LLMs using Hugging Face; exploring	17 Sep	7:00 pm	9:00 pm	9:00 pm - 9:15 pm	Rajdeep Ghosh (IIT KGP)	Same as above

	embeddings and tokenizers						
2	Advanced Prompt Engineering: CoT, Tree of Thought, auto-prompt optimization, and adversarial prompting	23 Sep	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Dr. Koustav Rudra (IIT KGP)	Prompting with open-source LLMs (Theory + Lab) (Llama2 / Mistral)
2	Hands-on Lab: Advanced Prompt Engineering with OpenAI GPT	24 Sep	7:00 pm	9:00 pm	9:00 pm - 9:15 pm	Rajdeep Ghosh (IIT KGP)	Prompting with open-source LLMs (Theory + Lab) (Llama2 / Mistral)
3	Multimodal GenAI: Text, Image, and Code using Diffusion, Stable Diffusion, DALL-E, and Codex	30 Sep	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Dr. Mainack Mondal (IIT KGP)	DeepLearning.AI – ChatGPT Prompt Engineering for Developers (Andrew Ng & Isa Fulford)
3	Hands-on Lab: Creative AI Lab on generating images with Stable Diffusion	1 Oct	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Nilanjan Sinhababu (IIT KGP)	https://www.deeplearning.ai/courses/prompt-engineering-for-vision-models
4	Hands-on Lab: Auto-code lab with Codex and Claude	7 Oct	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Nilanjan Sinhababu (IIT KGP)	https://www.youtube.com/watch?v=px7XlbYgk7I
4	Retrieval-Augmented Generation: Search,	8 Oct	6:00 pm	8:00 pm	8:00 pm -	Dr. Pawan Goyal (IIT KGP)	https://docs.langchain.com/oss/python/langchain/overview

	Retrieval, and Ragas Metrics				8:15 pm		
5	Hands-on Lab: Implementing RAG using LangChain with FAISS or ChromaDB	14 Oct	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Nilanjan Sinhababu (IIT KGP)	Hands-on session on LLM Finetuning and Agentic Workflows (Lab) Part - 1
5	Advanced NLP and Ethical AI: Sentiment Analysis and Explainable AI	15 Oct	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Dr. Mainack Mondal (IIT KGP)	- Introduction to NLP (Theory)
6	Hands-on Advanced NLP Lab: Sentiment analysis on customer-feedback datasets	28 Oct	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Rajdeep Ghosh (IIT KGP)	- Hands-on NLP (Lab)
6	Hands-on Explainable AI Lab: Practical application of LIME and SHAP	29 Oct	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Dr. Sabyasachi Mukhopadhyay (IIT KGP)	https://www.ibm.com/think/topics/explainable-ai
7	Agentic AI: ReAct, LangGraph, inference stack, MCP, and operational readiness	3 Nov	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Prof. Amlan Chakrabarti (IIT KGP)	https://arxiv.org/abs/2210.03629
7	Hands-on Agentic AI Lab: Reasoning core, tool integration, memory, and operations readiness	4 Nov	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Nilanjan Sinhababu (IIT KGP)	- Hands-on session on LLM Finetuning and Agentic Workflows (Lab) [agentic workflow part only]

8	Hands-on ROI Layer Lab: Guardrails, evaluations, and LLMOps in agents	12 Nov	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Rajdeep Ghosh (IIT KGP)	https://openai.github.io/openai-agents-python/guardrails/
8	Agentic Use Cases: Deep dive into agentic e-commerce architectures	13 Nov	6:00 pm	8:00 pm	8:00 pm - 8:15 pm	Prof. Amlan Chakrabarti (IIT KGP)	https://openai.com/business/guides-and-resources/a-practical-guide-to-building-ai-agents/